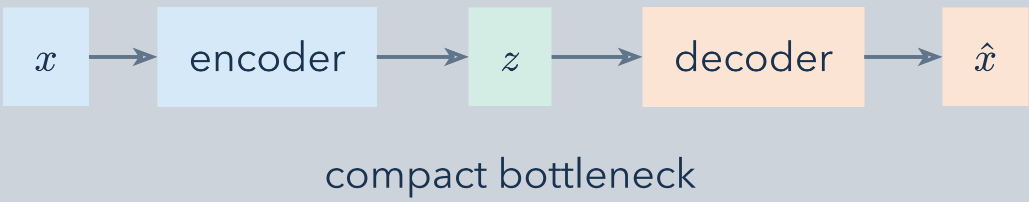
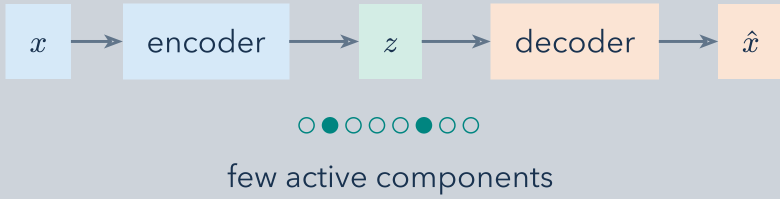
Every autoencoder learns to reconstruct an input through an intermediate representation. Compare the changing constraint or architecture while the common data path stays fixed.

1 Autoencoder



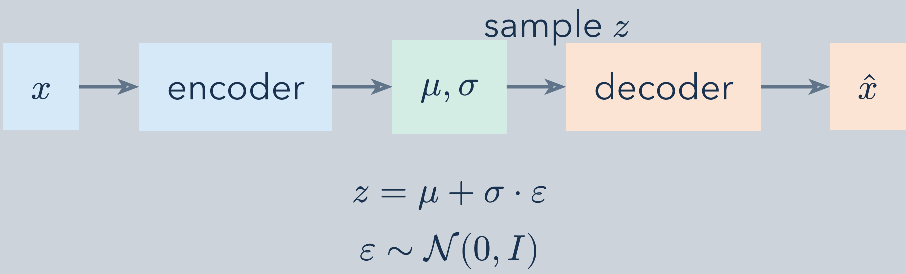
Encode x into a bottleneck z, then reconstruct $\hat{x}$. A reconstruction loss rewards retaining useful information.

2 Sparse autoencoder



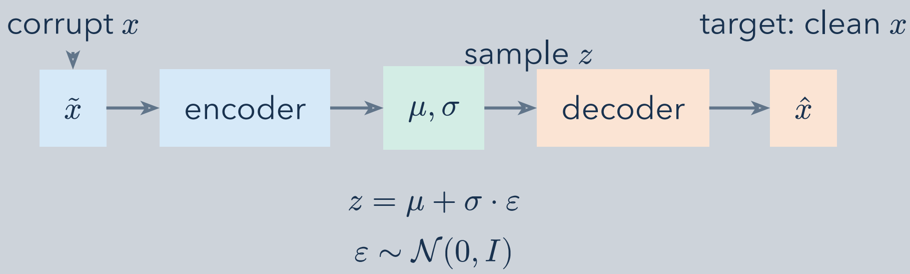
Penalize latent activity so only a few components activate for one input. The latent layer may be wider than the input; sparsity is the constraint.

3 Variational autoencoder



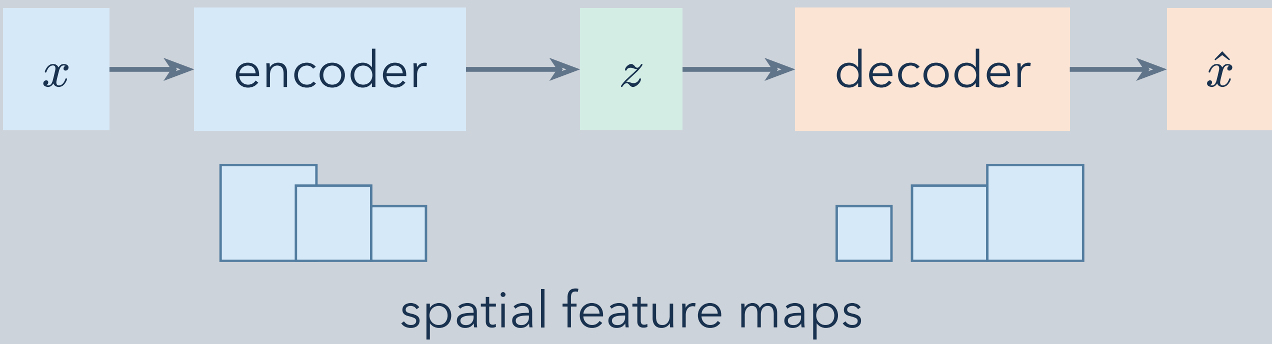
The encoder predicts a distribution over z, not just one code. Reconstruction and a KL-divergence term train it toward an explicit prior.

4 Denoising variational autoencoder



Corrupt the input, then reconstruct the clean target. The stochastic latent step remains; robustness and distribution learning are separate ingredients.

5 Convolutional autoencoder



Convolutions share filters over spatial locations. Feature maps change resolution through the encoder and decoder; this is an architectural choice.

6 These choices can be combined

representation

bottleneck · sparse · stochastic

architecture and training

convolutional · denoising

A convolutional model can also be sparse, variational, or denoising. These names describe different design dimensions, not mutually exclusive model families.

Notation: x = input, $\hat{x}$ = reconstruction, z = latent code, μ and σ = latent mean and standard deviation. The sampled noise ε enables the reparameterization step.